

High-Frequency of Groundwater Fluctuations, Underground Erosion, Potential Sinkhole Occurrences Across Long Island, NY

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Outline:

- Introduction and hypothesis
- Methodology: mapping past sinkhole formations on LI & GWL analysis
- Results: Figure of groundwater stations exhibiting high-frequency GWL fluctuations
- Discussion: speculation of the relationship between past sinkhole occurrences, water table depth, geological structures, and GWL oscillations.
- Conclusion

Introduction

- LI groundwater is located in aquifers; Upper Glacial aquifer is closest to surface
- Groundwater level (GWL) fluctuations are influenced by natural forces and human activities
- Underground erosion is a result of groundwater flow through porous sediments
- Sinkholes form when the surface above the underground erosion is destabilized and collapses inwards, this poses an environmental hazard
- Previous studies on sinkhole formation on LI have not been widely investigated

HYPOTHESIS: Groundwater stations exhibiting high frequencies of GWL oscillations is an identifier of accelerated underground erosion, and thus an area prone to occurrences of sinkhole formations across Long Island, New York.

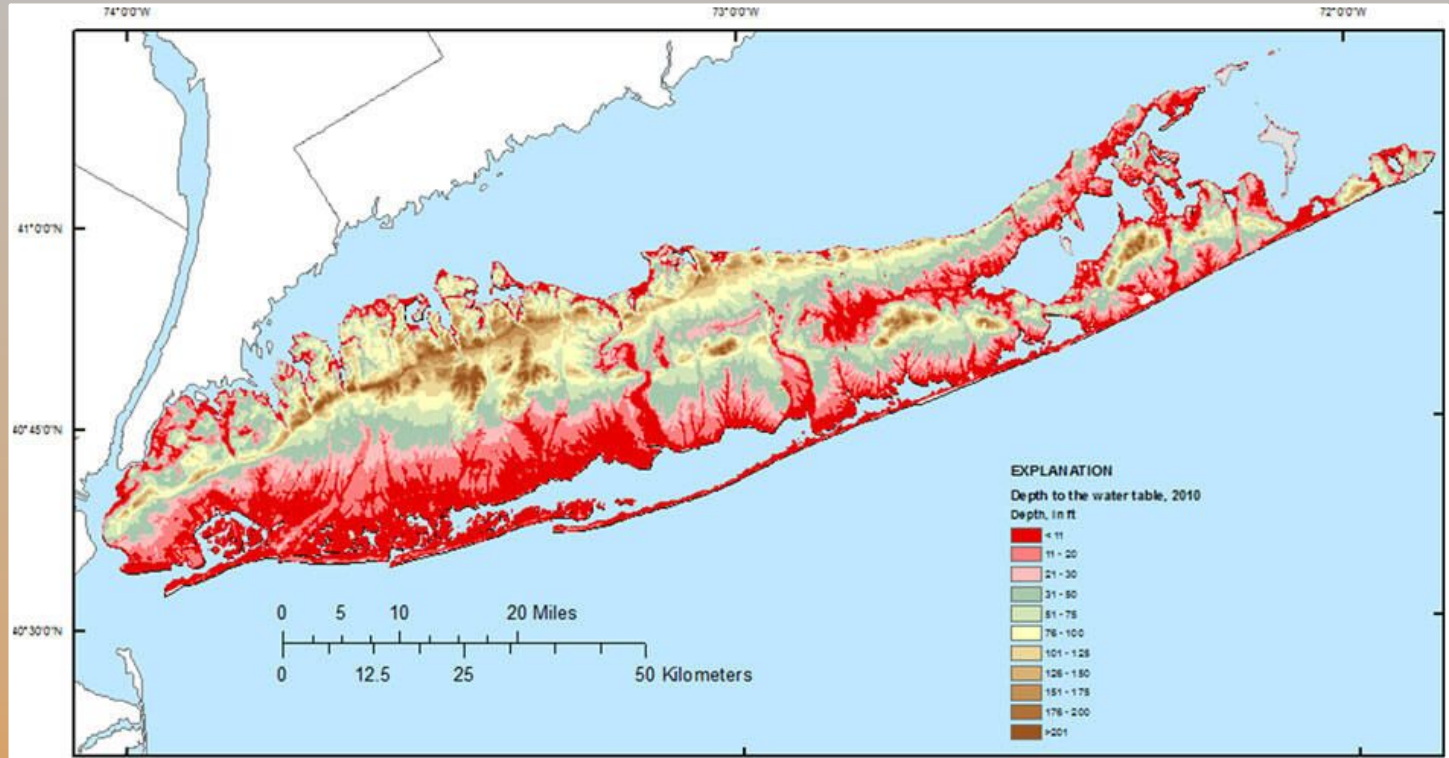


Figure 1. USGS map of the depth to the water table in feet on Long Island, NY 2010 (Como et al. 2013).

Sinkhole	Longitude	Latitude	Year	Depth (ft)
1) Huntington Station	-73.411268°	40.853503°	2023	6
2) Bay Ridge	-74.018219°	40.633155°	2024	20
3) Islip	-73.235022°	40.740219°	2014	12
4) Elmont	-73.712909°	40.700940°	2023	9
5) West Hempstead	-73.650142°	40.704847°	2023	6
6) Baldwin	-73.610711°	40.671756°	2023	n/a
7) Oceanside	-73.633872°	40.645077°	2023	20
8) Lido Beach	-73.629828°	40.587992°	2023	20

Table 1. Sinkhole formations on Long Island 2013-2023 and their respective locations and depths.

Methodology

- GWL data was downloaded from 58 USGS stations across LI
- Map of sinkhole formations on LI was created in Google Earth Pro
- GWL oscillations were analyzed in R using the following statistical tools:
 - Moving average
 - KZ filter from the kza package
 - Periodogram time-series analysis (TSA)
- Map of stations with high-frequency GWL fluctuations were mapped using the following R packages:
 - sf
 - ggspatial
 - ggplot2
 - Celestial

Results

- Sinkholes mostly occur on the southwestern parts of LI with some exception
- We observe a moderate spatial correlation between sinkhole occurrence and USGS stations showing high-frequency GWL oscillations

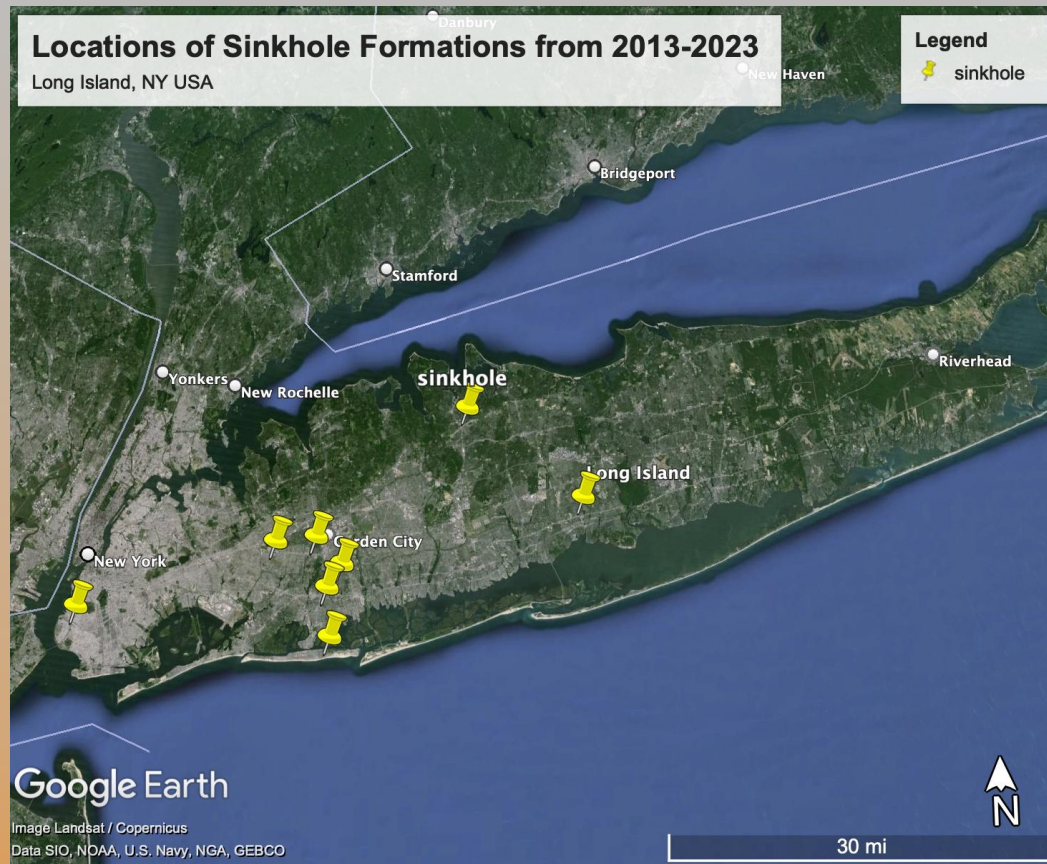


Figure 2. A Google Earth map showing the locations of sinkhole formations from 2013-2023 on Long Island, NY. Location data was sourced from local news reports.

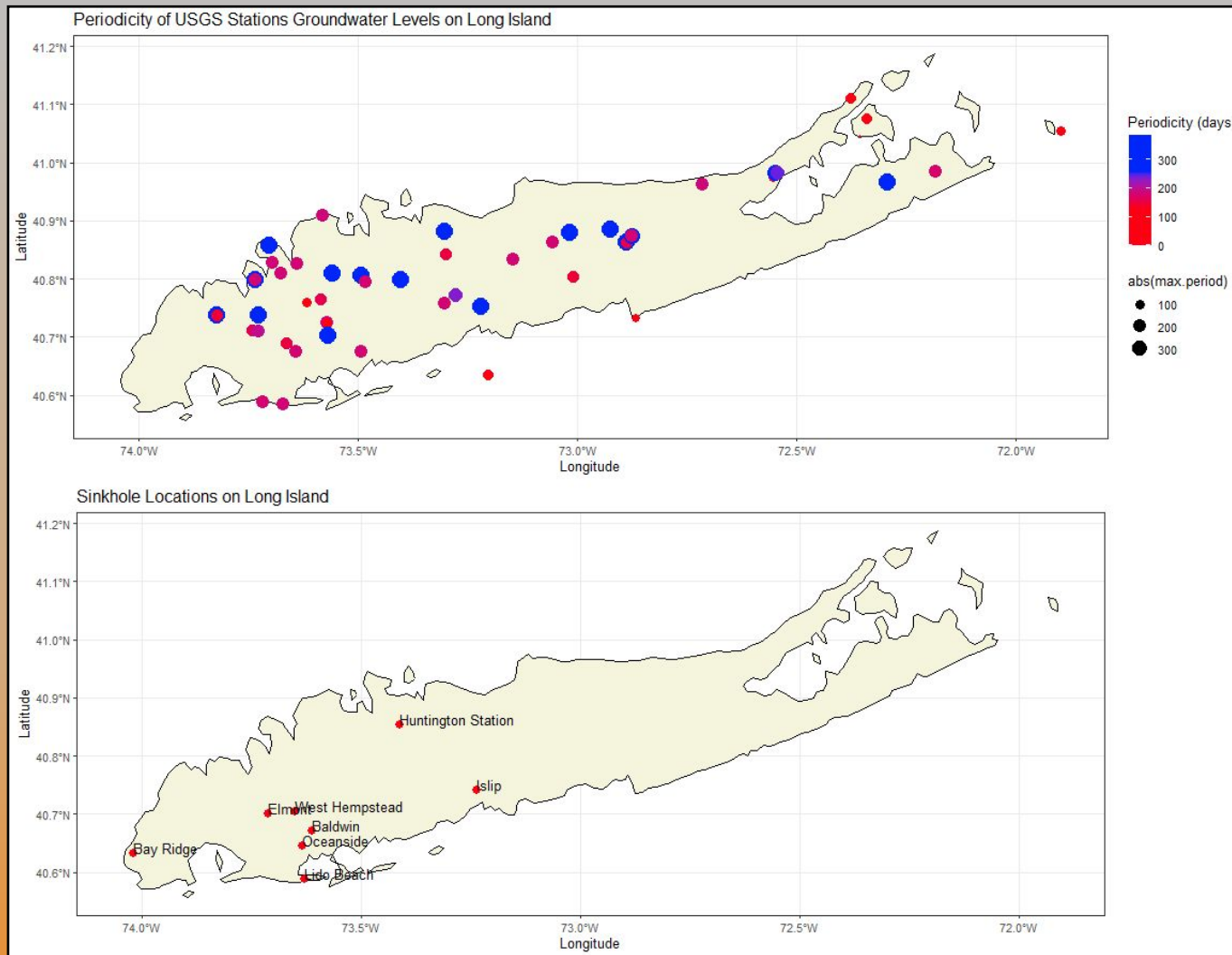


Figure 3. (Top) A GIS map showing the periodicity in days of USGS stations GWL across Long Island, NY. Higher frequency of GWL oscillations indicate lower periodicity (red dots). (Bottom) A map of the sinkhole locations from Table 1 across Long Island, NY.

Discussion

- Figure 1, 2, &3: Sinkhole formations are located at shallow depths to the water table, < 30 ft below surface.
- Figure 2: observed a slight linear pattern of sinkhole occurrences in Nassau County. Lack of previously mapped geological structures suggest sinkhole formation may be more anthropogenically influenced.
- Figure 3: Sinkhole formations on LI are moderately correlated with USGS station sites showing high-frequency GWL oscillations. Low periodicity indicates higher frequency of GWL fluctuations and higher frequency of oscillations.

Conclusion

- Identified a distinct relationship between shallow depth to water table and sinkhole formation occurrences
- High-frequency GWL oscillations occur at shallower depths to the water table on LI. These areas should be marked for further study due to any climatic conditions influencing GWL fluctuations to prevent flooding or sinkhole hazards.
- Speculation on the influence of adjacent fault regions near LI on underground erosion and sinkhole formation needs further studies
- Sinkhole occurrences are moderately related to high-frequency GWL oscillations at the USGS stations on LI.
- We can predict the relative location of future sinkhole formations by looking at the USGS stations showing high frequency GWL oscillations.
- Future studies would yield more coherent results for our hypothesis.

References

- Close B., Zurbenko I., Sun M. (2020). kza: Kolmogorov-Zurbenko Adaptive Filters. R package version 4.1.0.1, <https://CRAN.R-project.org/package=kza>
- Cohen, Phoebe. (2014) “Geological History of Long Island I.” *Williams Sites*, 15 March 2014, <https://sites.williams.edu/geos101/mid-atlantic/geological-history-of-long-island/>
- Como, M.D. et al. (2013). “Water-table and Potentiometric-surface altitudes in the Upper Glacial, Magothy, and Lloyd aquifers beneath Long Island, New York, April-May 2010” *New York Water Science Center*, [10.3133/sim3270](https://www.nywater.usgs.gov/archived_files/projects/gisunit/Long_Island_SIM3066_Handout.pdf)
- Dunnington D. (2023). _ggspatial: Spatial Data Framework for ggplot2_. R package version 1.1.9, <https://CRAN.R-project.org/package=ggspatial>
- Kennedy, Sean. “Three sinkholes emerge in two months | Herald Community Newspapers.” *LIHerald*, 7 August 2023, <https://www.liherald.com/stories/three-sinkholes-emerge-in-two-months,187898>. Accessed 2 March 2024.
- Krauth, Dan. “Repairs underway after broken sewer line causes 20-foot-deep sinkhole in Bay Ridge, Brooklyn.” *ABC7 New York*, 2 January 2024, <https://abc7ny.com/20-foot-sinkhole-bay-ridge-brooklyn-broken-sewer/14260368/>. Accessed 29 March 2024.
- Matthew C.D, Adam M. M.(2018), “Sinkhole formation mechanisms and geostatistical-based prediction analysis in a mantled karst terrain”, CATENA, Volume 165, 2018, Pages 333-344, ISSN 0341-8162, <https://doi.org/10.1016/j.catena.2018.02.010>
- Monti, J. et al. (2006) “Water Table and Potentiometric Surface Altitudes in the Upper Glacial, Magothy, and Lloyd Aquifers beneath Long Island, New York, March-April 2006.” *New York Water Science Center*, https://ny.water.usgs.gov/archived_files/projects/gisunit/Long_Island_SIM3066_Handout.pdf.
- NBC New York. “Long Island Recovers After Storm Dumps Record-Setting Rain, Causes Flooding.” 14 August 2014, <https://www.nbcnewyork.com/news/local/storm-cleanup-islip-record-rainfall-long-island-bay-shore-suffolk-nassau-ny/>. Accessed 2 March 2024.
- News 12. “New sinkhole brings traffic to a halt on Oceanside roadway.” *News 12 - Long Island*, 30 July 2023, <https://longisland.news12.com/new-sinkhole-brings-traffic-to-a-halt-on-oceanside-roadway>. Accessed 29 March 2024.
- Pebesma, E., Bivand, R. (2023). Spatial Data Science: With Applications in R. Chapman and Hall/CRC. <https://doi.org/10.1201/9780429459016>
- Robotham, A. (2018). “celestial: Collection of Common Astronomical Conversion Routines and Functions.” R package version 1.4.6, <https://CRAN.R-project.org/package=celestial>.
- Roje-Bonacci, T. (1997). “Influence of the fluctuation of groundwater levels upon the formation of sinkholes.” In *Engineering, Geology and the Environment*, Marinou PG (ed.). Proceedings of the International Symposium on Engineering Geology and the Environment 23-27 June, Athens, Greece. A. A Balkema: Rotterdam 997-1002
- Seelig, Jenn. “‘I was so scared.’ Mother and daughter wake up to massive sinkhole in West Hempstead yard.” *News 12 - Long Island*, 28 September 2023, <https://longisland.news12.com/i-was-so-scared-mother-and-daughter-wake-up-to-massive-sinkhole-in-west-hempstead-yard>. Accessed 2 March 2024.
- Tamiru A., Khuliso M., Haile M., Molla D. (2018) “Understanding the groundwater-level fluctuations for better management of groundwater resource: A case in the Johannesburg region, Groundwater for Sustainable Development”, Volume 7, 2018, Pages 1-7, ISSN 2352-801X, <https://doi.org/10.1016/j.gsd.2018.02.004>.
- Thorn, Kristen. “3 rescued from sinkhole outside home in Huntington Station, Long Island - ABC7 New York.” *abc7NY*, 26 January 2023, <https://abc7ny.com/sinkhole-rescue-huntington-station-long-island/12738556/>. Accessed 29 March 2024.
- Wickham, H. (2016) ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag New York, 2016. <https://ggplot2.tidyverse.org>